

UNIVERSITY COLLEGE OF JAFFNA

UNIVERSITY OF VOCATIONAL TECHNOLOGY

HIGHER NATIONAL DIPLOMA IN INFORMATION AND COMMUNICATION TECHNOLOGY

NATIONAL VOCATIONAL QUALIFICATION LEVEL 06

Initial Project Proposal
Online Food Ordering and Management System

Software Change Management (K72C001M12)

Assessment No 01

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Online Food Ordering and Management System

1. Project Description

The Online Food Ordering and Management System is a web-based system that allows customers to browse food items, view prices, select meals, and place food orders online.

Restaurants can use the system to manage their food menu, prices, availability, and customer orders. Customers can create accounts, select food items, add them to a shopping cart, provide delivery details, and place orders.

The main purpose of this system is to provide a simple and convenient way for customers to order food online while helping restaurants manage their menus and orders more efficiently.

2. Problem Statement

Many small and medium-sized restaurants still receive food orders through phone calls, social media messages, or directly at the restaurant. Managing these orders manually can cause problems such as incorrect orders, missing customer details, difficulty tracking orders, and delays in responding to customers.

Customers may also have difficulty viewing the complete menu, checking food availability, comparing prices, and knowing the current status of their orders.

Therefore, an online food ordering system is needed to provide a centralized platform where customers can view menus and place orders while restaurants can manage food items and customer orders in an organized way.

3. Project Objectives

The main objectives of the Online Food Ordering and Management System are:

- To develop a web-based platform for ordering food online.
- To allow customers to view available food items and prices.
- To allow customers to place food orders through the website.
- To provide a shopping cart for managing selected food items.
- To allow customers to manage their personal and delivery information.
- To help restaurants manage food items, prices, and availability.
- To help restaurants receive and manage customer orders.
- To allow customers to track their order status.
- To reduce errors associated with manual food ordering.
- To improve the efficiency of restaurant order management.

4. Target Users

The main users of the proposed system are:

1. Customers

Customers will be able to:

- Register and log in.
- View the restaurant menu.
- Search for food items.
- View food prices and descriptions.
- Add food items to the cart.
- Place food orders.
- Provide delivery details.
- View order history.
- Check order status.

2. Restaurant Staff

Restaurant staff will be able to:

- Log in to the system.
- Add new food items.
- Update food information.
- Change food prices.
- Manage food availability.
- View customer orders.
- Update order status.
- Manage customer order information.

3. Administrator

The administrator will be able to:

- Manage customer accounts.
- Manage restaurant staff accounts.
- Manage food categories.
- Manage food items.
- Manage orders.
- Monitor system activities.
- Generate basic reports.

5. Main System Features

User Registration and Login

- Customer registration
- Customer login
- Restaurant staff login
- Administrator login
- Logout
- User profile management

Food Menu Management

- Add food items
- Edit food items
- Delete food items
- Add food categories
- Update food prices
- Update food availability
- Add food descriptions and images

Food Search

- Search food items
- Filter food by category
- View food details
- View food price and availability

Shopping Cart

- Add food to cart
- Remove food from cart
- Change food quantity
- Calculate total price
- View selected food items

Online Ordering

- Place food orders
- Enter customer delivery details
- Confirm orders
- Generate order details
- View previous orders

Order Management

- View customer orders
- Accept orders
- Update order status

- Mark orders as preparing
- Mark orders as ready
- Mark orders as delivered
- View order history

Payment Management

- Select payment method
- Record payment information
- Display payment status
- Generate order receipt

For the initial version, payment can be implemented as a payment-record system or Cash on Delivery. A real payment gateway can be added later if required.

Administrator Management

- Manage users
- Manage food categories
- Manage food items
- Manage orders
- Monitor system activities
- Generate basic reports

6. Proposed Technology Stack

Frontend

HTML5

HTML will be used to create the structure of the website, including pages such as the home page, menu page, login page, cart page, and order page.

CSS3

CSS will be used to design the website, including layout, colors, fonts, buttons, forms, navigation, and responsive page design.

Backend

PHP

PHP will be used for the main server-side development of the online food ordering website. It will handle user accounts, food management, shopping carts, orders, and communication with the database.

C#

C# will be used for a supporting administration and reporting component where required. It can be used to create an administrative/reporting utility for generating and managing selected system reports.

Database

MySQL

MySQL will be used to store and manage system data, including:

- Customer information
- Administrator information
- Restaurant staff information
- Food categories
- Food items
- Shopping cart information
- Customer orders
- Order details
- Payment records
- Delivery information

Development Tools

- Visual Studio Code
- Visual Studio
- XAMPP
- MySQL

7. Expected Outcome

At the end of the project, a functional **Online Food Ordering and Management System** is expected to be developed.

The system will allow customers to register, browse food items, search the menu, add items to a shopping cart, place food orders, provide delivery information, and view their order status.

Restaurant staff will be able to manage food items, prices, availability, and customer orders. The administrator will be able to manage users, food categories, orders, and system information.

The proposed system is expected to reduce manual ordering problems, improve order management, provide customers with a convenient way to order food online, and help restaurants manage their daily food ordering activities more efficiently. The project will also provide practical experience in **HTML, CSS, PHP, C#, database management, web development, and system administration**.

8. Conclusion

The **Online Food Ordering and Management System** is proposed to provide a simple and efficient way for customers to order food online and for restaurants to manage their food items and customer orders. The system will reduce problems caused by manual ordering, improve order management, and provide customers with easier access to food menus and order information.

The proposed system will include customer management, food menu management, shopping cart, online ordering, order tracking, payment management, and administrator functions. The system will be developed using **HTML, CSS, PHP, C#, and MySQL**. At the end of the project, a functional web-based food ordering system is expected to be developed that can improve the overall food ordering and management process.